

ChronoFork 🕒 ↗️:

Learner—Multi-Agent Co-Roleplay for Divergent Historical Experiences

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app.chronofork.me

<https://github.com/IS492-SP26/team-project-chronofork-history-rewrite-simulation>



1. Problem & Target Users

✗ ChatGPT / Claude — great at subtasks

- But can't create a learning EXPERIENCE

👥 Who we're building for (after formative)

- Broader Population (e.g., History hobbyists)
- Not formal classrooms

🎯 Success — vs YouTube video/Digital Narrative

- More Interesting Experience
- Better Knowledge Retention
- Deeper Think (Context & Critical Thinking)

2. System Walkthrough



Scenario



Same event + same prompts

Config UI
(Panel Wizard)



Backend
(FastAPI+LLM)



Frontend
(Next.js)



Stage 1: Observe

Stage 2: Intervene

Stage 3: Reflect

2. System Walkthrough



Deployed:
config.chronofork.me

ChronoFork · Configuration

Stage 1: Configuration

Theme Input

E.g., Ancient China, WWII, The French Revolution...

Analyze Theme

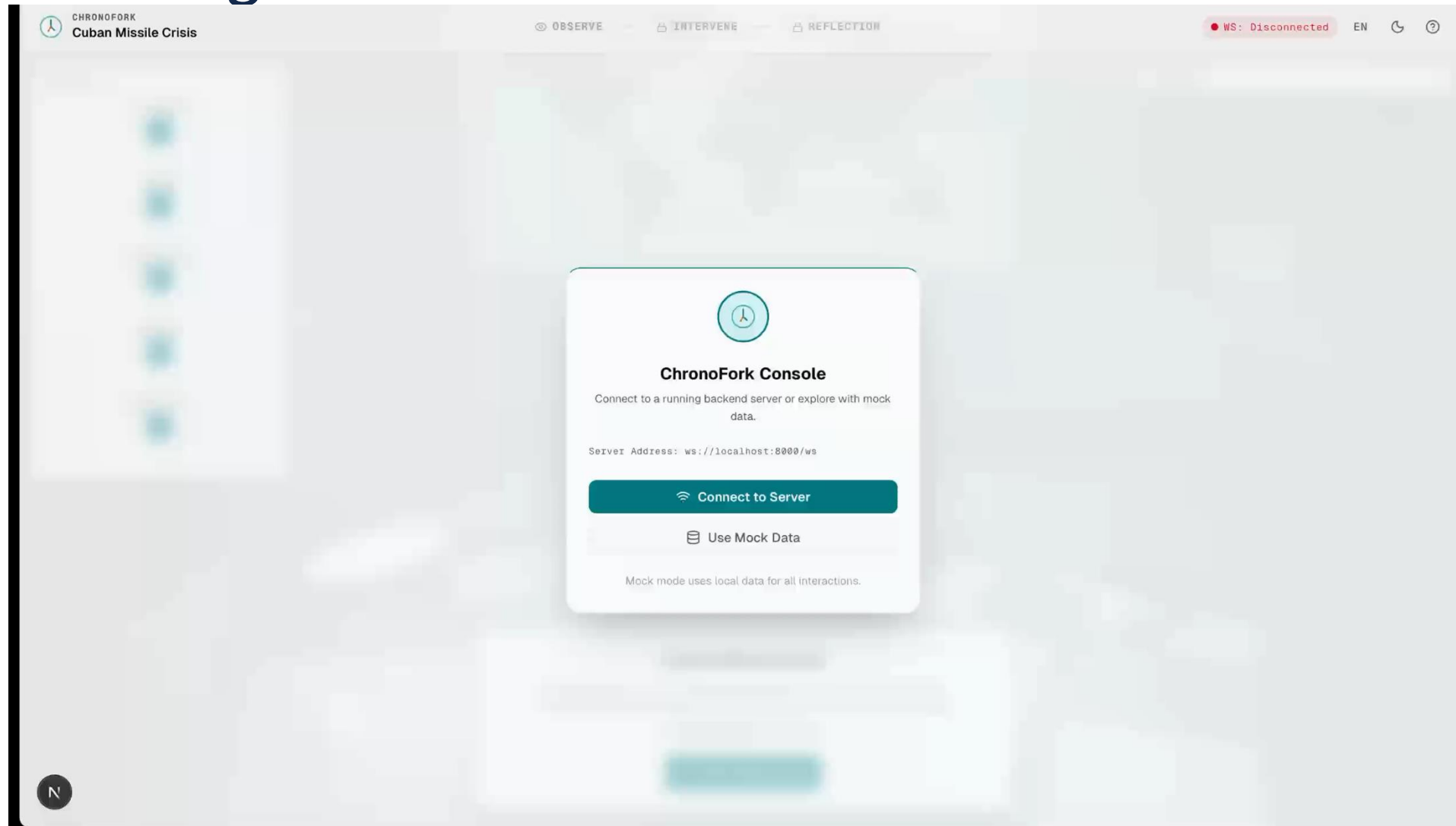
Stage 2: Cast

Stage 3: Storyline

2. System Walkthrough



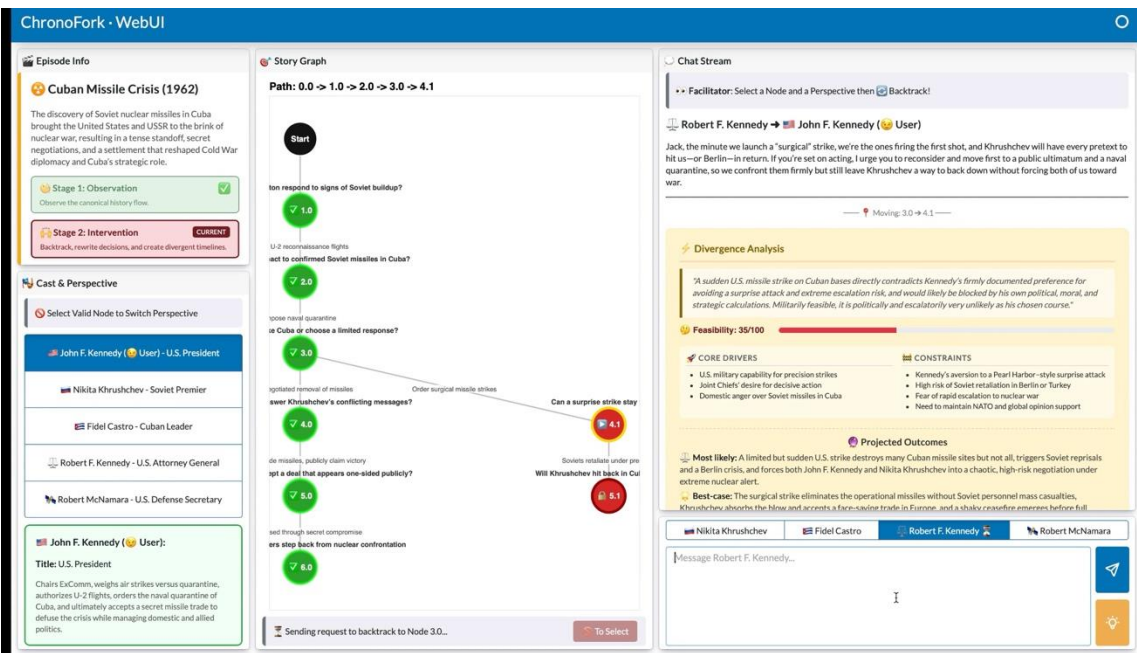
Deployed:
app.chronofork.me



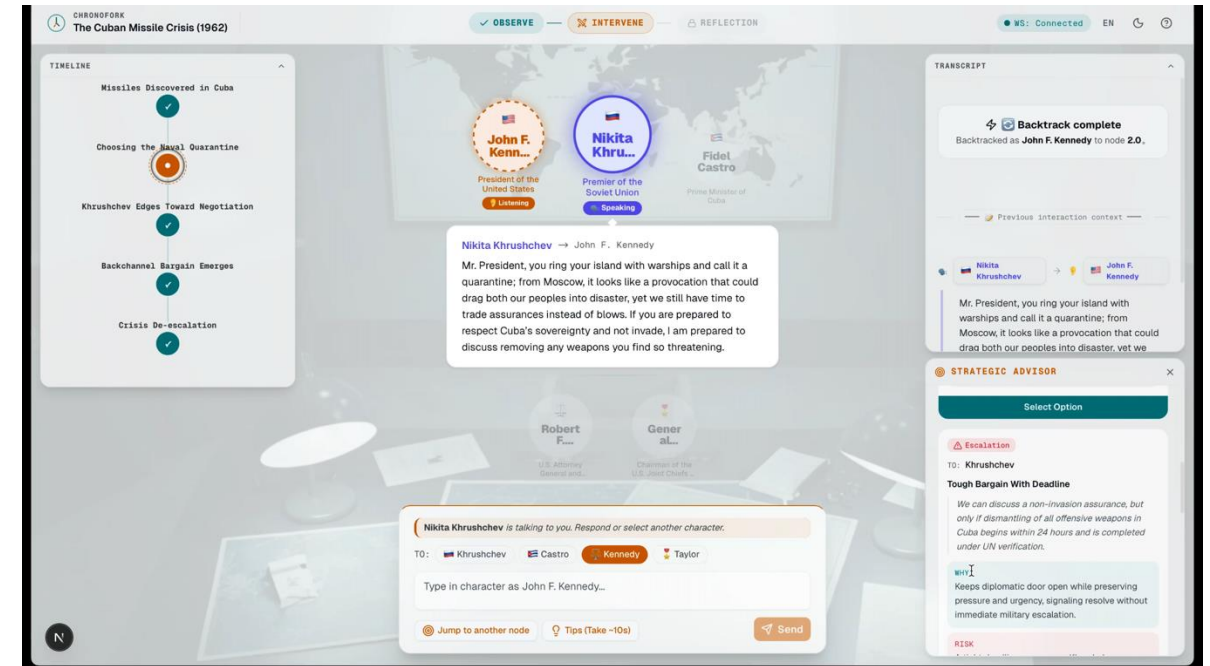
3. Design Decisions



Evolving Interface



Legacy
(webui.chronofork.me)



Current
(app.chronofork.me)

3. Design Decisions



Evolving Interface



VR
(Quest3)

Stronger immersion
but the scope was too wide,



keep web as the primary target
left VR as a separate thread.

3. Design Decisions



Responsiveness

User

[John F. Kennedy → Nikita Khrushchev]: Premier Khrushchev, unless you agree within 48 hours to remove all offensive weapons and personnel from Cuba, I am prepared to order a full invasion to finish what these strikes have begun.

Assistant 

`<meta targetName="John F. Kennedy" nodeid="end" />` You have already killed Soviet personnel and struck our sites; if you now invade Cuba, my generals will demand we answer in Berlin or against your missiles in Turkey, and no leader in Moscow could resist that pressure without losing all authority. Halt your ground plans and open a back channel at once, and I will order no retaliation beyond Cuba while we negotiate a rapid, supervised withdrawal of our remaining forces.



4. Safety

Privacy

- User messages → OpenAI API only

Jailbreak (after formative)

- Agents anchored to historical personas via system prompt
 - — narrative context channels input strongly
- Plausibility check gates divergence: "Use alien technology"
 - → rejected, no branch created

Reliability

- LLM responses cached → identical prompts = same demo
- 25 passing unit tests

5. Next Steps: User Study Design

Goal: Does ChronoFork improve historical understanding

- Participants: Broader Population (e.g., History hobbyists)

Design: Within-subjects, 3 conditions

- A Watch a documentary video (same event, ~10 min)
- B Interactive Digital Narrative (IDN)
- C ChronoFork session (Observe → Intervene → Reflect)

Measures:

- Knowledge retention, Critical thinking, Experience



6. Appendix

Contributions

- Individual Project
 - Implemented by Ziyi, Discussed with external collaborators & TA

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